

An Evaluation of Data Mining Models for Customer Churn Prediction

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INTRODUCTION

Customer churn refers to the situation in which customers discontinue their relationship with a company. In competitive industries such as telecommunications, customer retention is a critical business concern, as acquiring new customers is often more costly than retaining existing ones. As a result, the ability to identify customers who are likely to churn at an early stage is essential for organizations seeking to reduce revenue loss and improve long-term profitability.

With the increasing availability of customer-related data, data mining techniques have been widely applied to support predictive decision-making. Classification-based data mining models are commonly used to analyze customer behavior and predict churn by learning patterns from historical data, including service usage, contract details, and billing information. These models enable organizations to move beyond descriptive analysis and toward more proactive customer management strategies.

Despite their potential, churn prediction tasks present several challenges. One of the most significant challenges is class imbalance, where churned customers typically represent a much smaller proportion of the dataset compared to non-churn customers. In such cases, traditional evaluation metrics such as "accuracy" can be misleading. A model that predicts only the majority class may achieve high accuracy while failing to correctly identify customers who are at risk of churning. Therefore, performance metrics that focus on the minority class, such as precision, recall, F1-score, and ROC-AUC, are more appropriate for evaluating churn prediction models.

This project focuses on the evaluation of data mining classification models for customer churn prediction using the Telco Customer Churn dataset. Rather than proposing new algorithms, this work aims to compare baseline data mining models and examine the effects of commonly used improvement techniques, including imbalance handling strategies and feature representation approaches. By analyzing both baseline performance and model refinements, this project seeks to highlight the strengths and limitations of classical data mining methods when applied to churn prediction problems.

The remainder of this report is organized as follows. The next section presents the problem statement and research objectives. This is followed by a detailed description of the methodology, including dataset characteristics, data preprocessing steps, model selection, and experimental setup. Finally, the evaluation criteria are discussed to justify the performance metrics used in assessing model effectiveness for imbalanced churn prediction.

PROBLEM STATEMENT

Customer churn prediction is a challenging data mining task due to the complex nature of customer behavior and the characteristics of real-world business datasets. Although customer data often contains rich information related to services, contracts, and billing patterns, the relationships between these attributes and churn outcomes are not always explicit or linear. This makes it difficult to construct models that can reliably distinguish between customers who are likely to churn and those who are not.

One of the primary challenges in churn prediction is class imbalance. In the Telco Customer Churn dataset, the number of non-churn customers significantly exceeds the number of churn customers. As a result, many classification models tend to favor the majority class during training. This imbalance can lead

to biased predictions in which churners are underrepresented, reducing the practical usefulness of the model for retention purposes.

Another key issue is the reliance on inappropriate evaluation metrics, particularly accuracy. In imbalanced datasets, accuracy can produce misleading conclusions, as a model that predicts only the majority class may still achieve a high accuracy score while completely failing to identify churn cases. For churn prediction, the cost of misclassifying a churner is typically higher than misclassifying a non-churner. Therefore, evaluation metrics that consider both false positives and false negatives are required to provide a more realistic assessment of model performance.

In addition, commonly used improvement techniques, such as oversampling methods and class balancing strategies, do not always guarantee better performance. While these techniques are often recommended in imbalanced classification problems, their effectiveness depends on the characteristics of the dataset and the underlying model. Applying such methods without proper evaluation may introduce noise or distort feature distributions, leading to reduced predictive performance.

Given these challenges, there is a need for a systematic evaluation of data mining classification models for customer churn prediction. This includes not only comparing baseline models but also examining the impact of different improvement strategies and evaluation metrics. Understanding the limitations and trade-offs of these approaches is essential for selecting models that are both technically sound and practically useful in churn prediction scenarios.

OBJECTIVES

The primary objective of this project is to evaluate the effectiveness of data mining classification models in predicting customer churn using the Telco Customer Churn dataset. This evaluation focuses on understanding model behavior in the presence of class imbalance and identifying suitable evaluation metrics for churn prediction tasks.

The specific objectives of this project are as follows:

1. To apply and compare multiple **baseline data mining classification models** for customer churn prediction using a consistent experimental setup.
2. To examine the impact of **class imbalance** on model performance and prediction outcomes.
3. To evaluate commonly used **model improvement techniques**, such as class weighting, oversampling methods, and feature representation strategies, in the context of churn prediction.
4. To assess model performance using appropriate evaluation metrics, including **precision, recall, F1-score**, and **ROC-AUC**, rather than relying solely on accuracy.
5. To identify the strengths and limitations of classical data mining approaches for customer churn prediction and provide insights into their practical applicability.

These objectives guide the experimental design and evaluation process of this project, ensuring that the analysis remains focused on both technical validity and real-world relevance.

DATASET DESCRIPTION

This project utilizes the **Telco Customer Churn dataset**, which contains customer-level information from a telecommunications service provider. The dataset is publicly available on Kaggle and is widely used in data mining and churn prediction studies. It provides a comprehensive view of customer demographics, service subscriptions, and billing-related attributes, making it suitable for analyzing churn behavior. The

dataset consists of **7,043 customer records**, with each record representing an individual customer. The target variable, "**Churn**", indicates whether a customer has discontinued the service. Churned customers form a minority class, resulting in a **class imbalance** that poses challenges for predictive modeling.

The attributes in the dataset can be broadly categorized into the following groups:

- **Demographic features**, such as gender, senior citizen status, and dependent information
- **Service-related features**, including internet service type, phone service, and additional service subscriptions
- **Contract and billing features**, such as contract type, payment method, monthly charges, and total charges

Most features are categorical in nature, while a smaller number of variables are numerical. This mixture of feature types requires careful preprocessing and feature representation to ensure compatibility with data mining classification models. Given its imbalanced nature and high proportion of categorical attributes, the Telco Customer Churn dataset provides an appropriate testbed for evaluating classical data mining models and examining the effectiveness of different preprocessing and evaluation strategies.

METHODOLOGY

This project follows a structured **data mining methodology** based on the "Knowledge Discovery in Databases (KDD)" process. The methodology consists of data preprocessing, feature representation, model selection, experimental setup, and evaluation. This structured approach ensures that the analysis is systematic, reproducible, and suitable for comparing multiple classification models under consistent conditions.

Data Preprocessing

Data preprocessing is a fundamental step in the data mining process, as raw data often contain inconsistencies and attributes that are not directly suitable for analysis. In this project, preprocessing includes **data cleaning**, **data transformation**, and **attribute selection**. Non-informative attributes, such as unique customer identifiers, are removed to prevent them from influencing model training.

The target variable, "**Churn**", is transformed into a binary label to support **supervised classification**. Missing or inconsistent values in numerical attributes are handled to ensure data quality and model stability. These preprocessing steps aim to reduce noise while preserving meaningful patterns that are relevant to customer churn behavior.

Feature Representation

Data preprocessing is a fundamental step in the data mining process, as raw data often contain inconsistencies and attributes that are not directly suitable for analysis. In this project, preprocessing includes **data cleaning**, **data transformation**, and **attribute selection**. Non-informative attributes, such as unique customer identifiers, are removed to prevent them from influencing model training.

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Model Selection

This project evaluates five **data mining classification models** that are commonly applied to binary classification and customer churn prediction problems. The selected models represent a diverse set of learning paradigms, enabling a comprehensive comparison across different modeling assumptions and decision mechanisms.

The five models examined in this experiment are:

- **Logistic Regression**, a linear classification model that estimates the probability of churn using a logistic function
- **Decision Tree**, a rule-based classifier that partitions data through hierarchical decision rules
- **k-Nearest Neighbors (k-NN)**, an instance-based learning model that classifies observations based on similarity to neighboring instances
- **Support Vector Machine (SVM)**, a margin-based classifier that seeks an optimal separating boundary between classes
- **Naive Bayes**, a probabilistic classifier based on Bayes' theorem and conditional independence assumptions

All five models are first evaluated as **baseline models** using consistent preprocessing and default parameter configurations to establish reference performance levels. Following baseline evaluation, each model is further refined using **model improvement techniques**, including parameter tuning and strategies designed to address class imbalance and feature representation challenges. This two-phase approach allows the study to assess both inherent model performance and the impact of systematic refinement.

Experimental Setup

The dataset is divided into **training** and **testing** subsets using a stratified sampling strategy to preserve the original distribution of churn and non-churn classes. This partitioning is applied consistently across all models to ensure fair and comparable evaluation.

The experimental workflow consists of two main phases:

1. **Baseline Phase**

In this phase, all five classification models are trained using the same training data and evaluated on the same testing data with default parameter settings. This phase establishes baseline performance metrics for comparative analysis.

2. **Improvement Phase**

In the improvement phase, all five models are refined using selected **model optimization techniques**. These techniques include **hyperparameter tuning**, as well as approaches for addressing **class imbalance** and improving feature representation. The refined models are evaluated using the same testing data and evaluation metrics as in the baseline phase, allowing direct comparison between baseline and improved configurations.

This experimental design ensures that observed performance differences are attributable to model choice and improvement strategies, rather than inconsistencies in data partitioning or evaluation methodology.

Evaluation Criteria

The performance of data mining classification models in this study is evaluated using metrics that are appropriate for **imbalanced binary classification** problems. Since customer churn represents a minority class in the dataset, relying solely on overall correctness can lead to misleading conclusions. Therefore, multiple evaluation metrics are employed to provide a comprehensive assessment of model effectiveness.

The evaluation metrics consist of these:

- **Accuracy** measures the proportion of correctly classified instances among all predictions. While accuracy is commonly reported in classification tasks, it is **not suitable as the primary evaluation metric** for churn prediction. In imbalanced datasets, a model that predicts only the majority class can achieve high accuracy while failing to identify churners. For this reason, accuracy is reported for completeness but is not used for model selection or comparison in this study.
- **Precision** measures the proportion of correctly predicted churn cases among all instances classified as churn. In the context of customer churn, precision reflects the **reliability of churn predictions**, indicating how many flagged customers are actual churners. A low precision value implies a high number of false positives, which may lead to unnecessary retention efforts.
- **Recall** measures the proportion of actual churn cases that are correctly identified by the model. High recall is particularly important in churn prediction, as failing to detect a churner may result in lost revenue. This metric reflects the model's ability to capture customers who are at risk of leaving.
- **F1-score** is the harmonic mean of precision and recall, providing a balanced measure that accounts for both false positives and false negatives. Since churn prediction requires a trade-off between identifying churners and avoiding excessive false alarms, the **F1-score is used as the primary metric** for comparing model performance in this study.
- **Receiver Operating Characteristic Area Under the Curve (ROC-AUC)** measures the model's ability to distinguish between churn and non-churn customers across different decision thresholds. A higher ROC-AUC value indicates stronger discriminative power and more reliable ranking of churn risk. This metric is particularly useful for assessing overall model quality independent of a fixed classification threshold.

In this study, **F1-score** and **ROC-AUC** are selected as the primary evaluation metrics, as they jointly capture decision-level performance and ranking capability in the presence of class imbalance, supported by precision and recall provide additional insight into model behavior. Accuracy is included for reference but is not used as a decision criterion due to the imbalanced nature of the churn dataset. This evaluation framework ensures that model performance is assessed in a manner that is both technically sound and practically relevant for churn prediction tasks.

EVALUATION AND DISCUSSION

This section presents the evaluation results of the five data mining classification models used in this study and discusses their performance under both baseline and improved configurations. The evaluation focuses on **F1-score** and **ROC-AUC** as the primary metrics, as these measures are more appropriate for imbalanced churn prediction tasks than accuracy.

Baseline Model Evaluation

This section discusses the baseline performance of the five data mining classification models using default configurations and without applying any model improvement techniques. The evaluation focuses on **accuracy**, **precision**, **recall**, **F1-score**, and **ROC-AUC**, with particular emphasis on F1-score and ROC-AUC due to class imbalance in the dataset. Here is the result:

	model	accuracy	precision	recall	f1	roc_auc
0	SVM	0.726929	0.490805	0.761141	0.596785	0.798994
1	Logistic Regression	0.718883	0.480747	0.734403	0.581100	0.805640
2	Random Forest	0.775674	0.576991	0.581105	0.579041	0.816221
3	XGBoost	0.772362	0.571174	0.572193	0.571683	0.812401
4	Decision Tree	0.709418	0.459909	0.541889	0.497545	0.657991

- **Support Vector Machine (SVM)**

The Support Vector Machine achieves an accuracy of **0.7269**, with a recall of **0.7611**, indicating that it is effective at identifying churn customers. However, its precision is relatively low at **0.4908**, meaning that a large proportion of customers predicted as churn do not actually churn. This imbalance between precision and recall results in an F1-score of **0.5968**, which is the highest among the baseline models. The ROC-AUC score of **0.7990** further suggests that SVM provides reasonable class separation at the baseline level

- **Logistic Regression**

Logistic Regression records an accuracy of **0.7189**, with a recall of **0.7344**, showing a strong ability to detect churners. Like SVM, its precision remains low at **0.4807**, indicating a high number of false positive churn predictions. The resulting F1-score is **0.5811**, slightly lower than that of SVM. However, Logistic Regression achieves a ROC-AUC of **0.8056**, demonstrating good ranking capability despite its simplicity as a linear model.

- **Random Forest**

Random Forest attains a relatively high accuracy of **0.7757**, which is the highest among all baseline models. Its precision of **0.5770** is higher than that of linear and margin-based models, indicating more reliable churn predictions. However, its recall drops to **0.5811**, suggesting that a significant portion of churn customers are not detected. This trade-off leads to an F1-score of **0.5790**. The ROC-AUC value of **0.8162** indicates strong overall discrimination capability, even though decision-level performance remains moderate.

- **XGBoost**

XGBoost produces an accuracy of **0.7724**, closely matching that of Random Forest. It achieves a precision of **0.5712** and a recall of **0.5722**, resulting in a balanced but moderate F1-score of

0.5717. The ROC-AUC score of **0.8124** suggests that XGBoost is effective at ranking customers by churn risk, even though its default threshold does not yield optimal decision-level performance in the baseline setting.

- **Decision Tree**

Decision Tree records an accuracy of **0.7094**, with a precision of **0.4599** and recall of **0.5419**. These values indicate weaker predictive performance compared to the other models. The resulting F1-score of **0.4975** is the lowest among the baseline models, reflecting limited generalization ability when using a single-tree structure. Its ROC-AUC of **0.6580** further confirms that the model struggles to separate churn and non-churn customers effectively at the baseline level.

Baseline Performance Summary

Overall, the baseline evaluation highlights clear trade-offs between accuracy, precision, and recall across models. While ensemble-based models such as Random Forest and XGBoost achieve higher accuracy and ROC-AUC, they exhibit lower recall under default settings. In contrast, SVM and Logistic Regression demonstrate stronger recall but suffer from lower precision. These results indicate that baseline configurations are insufficient for effective churn prediction and motivate the application of targeted model improvement techniques in the subsequent evaluation phase.

Improved Model Evaluation

This section presents the evaluation results after applying model improvement techniques to all five data mining classifiers. These improvements include hyperparameter tuning, class imbalance handling, feature refinement, and threshold optimization. Performance is assessed using the same evaluation metrics as in the baseline phase to ensure a fair comparison. Here is the result:

	model	accuracy	precision	recall	f1	roc_auc
0	Random Forest (Tuned CV, Balanced)	0.768102	0.542720	0.803922	0.647989	0.841789
1	XGBoost (Tuned CV, Balanced)	0.773781	0.553687	0.762923	0.641679	0.844303
2	Logistic Regression (Version 1 (Ridge + Poly 2))	0.783247	0.573257	0.718360	0.637658	0.832682
3	Decision Tree (Optimized)	0.793658	0.609457	0.620321	0.614841	0.830647
4	SVM (Improved)	0.789399	0.600346	0.618538	0.609306	0.815076

- **Random Forest (Tuned CV, Balanced)**

The improved Random Forest model achieves an accuracy of **0.7681**, with a substantial increase in recall to **0.8039**, indicating a strong ability to correctly identify churn customers. Although precision decreases slightly to **0.5427**, the improved balance between precision and recall results in the highest F1-score of **0.6480** among all models. The ROC-AUC value of **0.8418** further confirms robust discrimination capability. These results demonstrate that class balancing and hyperparameter tuning significantly enhance the model's effectiveness in churn prediction.

- **XGBoost (Tuned CV, Balanced)**

XGBoost records an accuracy of **0.7738**, with a recall of **0.7629** and a precision of **0.5537**. This balanced performance yields an F1-score of **0.6417**, placing XGBoost as the second-best performing model in terms of F1-score. Notably, XGBoost achieves the highest ROC-AUC score of

0.8443, indicating superior ranking ability in distinguishing churn and non-churn customers. These results suggest that XGBoost is particularly effective for prioritizing customers based on churn risk.

- **Logistic Regression (Version 1: Ridge + Polynomial Degree 2)**

The improved Logistic Regression model shows a notable performance increase, achieving an accuracy of **0.7832** and a precision of **0.5733**. With a recall of **0.7184**, the model maintains strong churn detection capability while reducing false positives. The resulting F1-score of **0.6377** reflects a meaningful improvement over the baseline configuration. A ROC-AUC of **0.8327** indicates that the model benefits from feature expansion and regularization, despite its linear foundation.

- **Decision Tree (Optimized)**

The optimized Decision Tree achieves an accuracy of **0.7937**, the highest accuracy among all improved models. Its precision of **0.6095** is also the highest observed, indicating reliable churn predictions. However, recall is lower at **0.6203**, which limits the model's ability to detect all churn cases. This trade-off results in an F1-score of **0.6148**. The ROC-AUC score of 0.8306 suggests moderate overall discrimination performance, with improvements primarily driven by reduced overfitting.

- **Support Vector Machine (Improved)**

The improved SVM achieves an accuracy of **0.7894**, with precision and recall values of **0.6003** and **0.6185**, respectively. These balanced but moderate values produce an F1-score of **0.6093**, representing an improvement over the baseline SVM. The ROC-AUC score of **0.8151** indicates stable ranking capability, although gains are less pronounced compared to ensemble-based models.

Improved Performance Summary

Overall, the improvement phase results in clear performance gains across all models when compared to their baseline counterparts. Ensemble-based models, particularly **Random Forest** and **XGBoost**, achieve the strongest F1-scores and ROC-AUC values, highlighting their suitability for churn prediction tasks involving imbalanced data. While simpler models such as Logistic Regression and Decision Tree benefit from targeted optimization, their performance remains constrained relative to ensemble methods. These findings confirm that systematic model refinement and appropriate imbalance handling significantly enhance churn prediction effectiveness.

CONCLUSION

This study evaluated the effectiveness of five data mining classification models for customer churn prediction using the Telco Customer Churn dataset. The evaluation was conducted through a structured methodology that included baseline model comparison and systematic model improvement, with particular attention to challenges arising from class imbalance.

The baseline results showed that while all models were able to capture general churn patterns, their performance was limited, especially in terms of balancing precision and recall. Models such as Support Vector Machine and Logistic Regression demonstrated high recall but suffered from low precision, while ensemble-based models achieved higher accuracy and ROC-AUC but initially failed to identify churners effectively.

After applying model improvement techniques, including hyperparameter tuning, class balancing strategies, and feature refinement, all five models exhibited performance improvements. Among them, **Random Forest** achieved the highest F1-score, indicating strong decision-level performance in identifying

churn customers. **XGBoost** demonstrated the highest ROC-AUC, highlighting its effectiveness in ranking customers according to churn risk. Simpler models such as Logistic Regression, Decision Tree, and Support Vector Machine also benefited from optimization, although their performance remained comparatively lower than ensemble-based approaches.

Overall, the results confirm that churn prediction is highly sensitive to evaluation metrics and imbalance handling strategies. Accuracy alone is insufficient for assessing model effectiveness in this context. Instead, metrics such as F1-score and ROC-AUC provide a more meaningful evaluation of churn prediction performance. This study demonstrates that careful model selection and targeted improvement techniques are essential for building effective data mining solutions for customer churn prediction.

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